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Article

Supplementary Material: WRF Cold-Pool Sensitivities Relevant to Winter-Ozone Modelling in the Uinta Basin

John R. Lawson [1,2,*], Michael J. Davies [1], Loknath Dhar [1,3] and Seth N. Lyman [1,3]

[1] Bingham Research Center, Utah State University, Vernal, UT 84078, USA;
[2] Department of Mathematics and Statistics, Utah State University, Logan, UT 84322, USA;
[3] Department of Chemistry and Biochemistry, Utah State University, Logan, UT 84322, USA
* Correspondence: john.lawson@usu.edu

1. Secondary Controlled Comparisons

Table 1 collects the secondary process and fine-terrain feedback responses on the common station and common-AGL pipeline.

Table 1. Mean secondary controlled contrasts over the five hourly spatial medians from 1400 to 1800 UTC. PBL height is scheme defined across the MYJ/Eta versus YSU/revised-MM5 comparison and is retained only with that caveat.

Metric	X1 − X0	X2 − X1	X9 − X0
T2 bias (°C)	+0.001	+0.37	-0.004
10 m wind absolute error (m s^{-1})	+0.001	-0.145	+0.03
Surface-pressure bias (Pa)	-0.12	+6.3	-0.77
Stability $S_{50:500}$ (K)	+0.004	+0.11	+0.04
Transport wind $U_{50:500}$ (m s^{-1})	+0.006	+0.098	-0.01
PBL height (m)	+0.01	-9.2	+0.05

Table 2 summarises the controlled terrain-source brackets without treating them as a source ranking.

Table 2. Terrain-source responses for the controlled NAM one-way brackets. The first operation is default minus fine terrain; the coarse-source operations use the deliberately coarse source as the second member.

Contrast	Terrain difference	2 m response	Scope
X3 − X0, default minus fine	7.9 m RMS; -55 to +60 m range	Elevation-band means 0.07–0.27 K; terrain correlation below 0.3 at 1400–1600 UTC	Small d02 response; insufficient for a terrain-source ranking
X3 − X4 and X0 − X4, coarse brackets	91.9/93.1 m RMS; extrema to -462/+487 m	Slope-band means 1.2–1.5 K; terrain correlation 0.50–0.67	Response varies with domain support and diagnostic

The controlled slope-aware-radiation treatment produced negligible response at the sampled basin-floor columns but an aspect-organised skin-temperature response on steep terrain. The archived shortwave field is a flat-plane quantity and did not express the

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treatment, while the archive lacks the surface heat-flux terms needed to test the mechanism directly. Figure 1 retains the spatial diagnostic with those limitations.

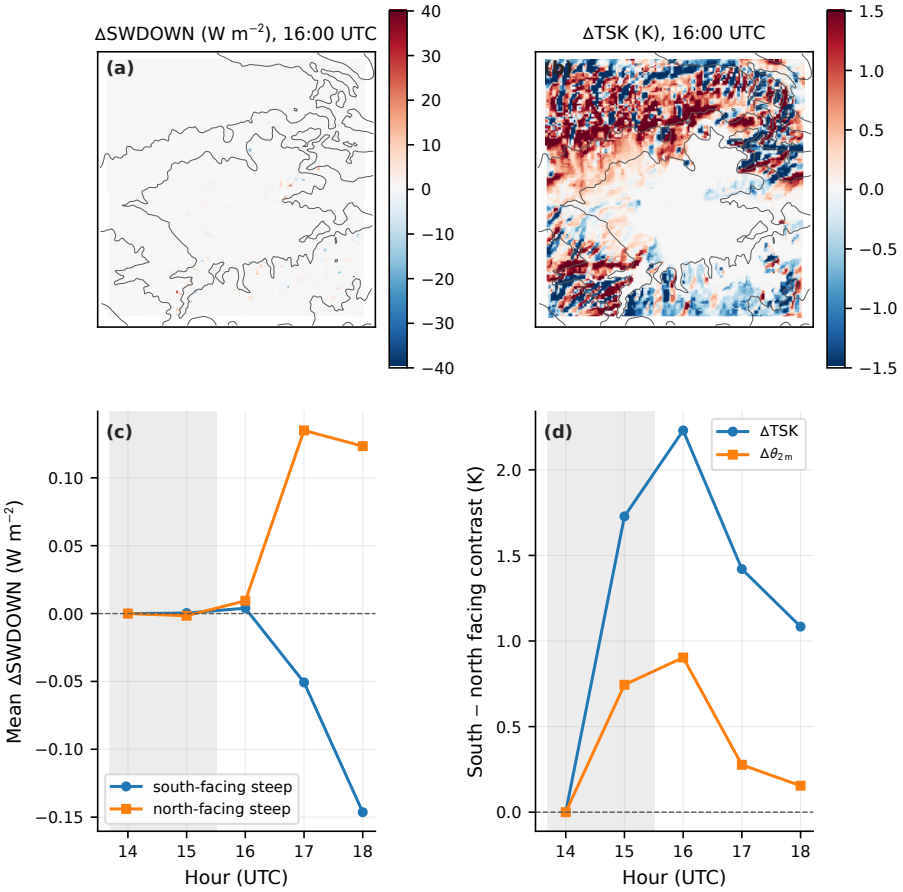


Figure 1. Diagnostic response to slope-aware radiation and terrain shading on the controlled fine-terrain branch. The archived flat-plane shortwave field showed negligible differences, whereas skin temperature showed an aspect-organised response on steep terrain. The figure demonstrates that the common station metric can miss a localised model response, but it does not show that slope-aware radiation improved or degraded the cold pool and supports no configuration ranking.